

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

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Answer

1. Introduction

Artificial Intelligence (AI) and Machine Learning (ML) are widely used to solve prediction problems in education, healthcare, finance, and many other domains. One important application of AI is predicting student scores based on factors such as attendance, study hours, previous academic performance, assignments, and classroom participation.

A **Neural Network** is a machine learning model inspired by the human brain. It learns patterns from training data and predicts outputs for new, unseen data. After training, the performance of the model is evaluated using error metrics such as **Mean Squared Error (MSE)**.

In the given problem, the neural network has:

- **Training MSE = 0.85**
- **Testing MSE = 5.20**

The training error is very small, but the testing error is much higher. This indicates that the model performs well on the training data but fails to generalize to new data. Such a situation usually indicates **overfitting**.

2. Neural Network Overview

A Neural Network consists of three main layers.

Input Layer

Receives the input features.

Example:

- Study hours
- Attendance
- Previous marks
- Assignment scores

Hidden Layer

Processes information using neurons, weights, biases, and activation functions.

Output Layer

Produces the predicted student score.

Example:

Input Layer

(Study Hours, Attendance, Previous Marks)



Hidden Layer 1



Hidden Layer 2



Output Layer

(Predicted Score)

The objective is to minimize prediction error during training.

3. Mean Squared Error (MSE)

Mean Squared Error is one of the most commonly used loss functions for regression problems.

Formula

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where

- **n** = Number of samples
- **y** = Actual score
- **$\hat{y}$** = Predicted score

The lower the MSE, the better the model performs.

Example:

Actual Score	Predicted Score	Error	Squared Error
80	82	2	4
75	74	-1	1
90	88	-2	4

Total Squared Error = 9

MSE = 9 / 3 = 3

A lower MSE means predictions are closer to actual values.

4. Analysis of the Given Problem

Given:

Training MSE = **0.85**

Testing MSE = **5.20**

Comparison:

Dataset MSE

Training 0.85

Testing 5.20

Interpretation

- The model performs extremely well during training.
- The model performs poorly during testing.
- The testing error is almost **6 times larger** than the training error.

This indicates that the neural network has **memorized the training data instead of learning general patterns.**

This problem is called **Overfitting.**

5. Understanding Overfitting

Overfitting occurs when the model learns unnecessary details, noise, or random patterns in the training data.

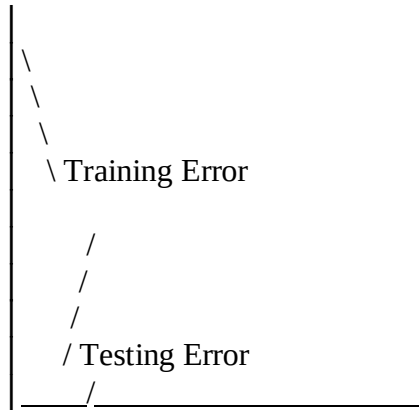
Instead of learning general relationships, it memorizes the dataset.

Characteristics:

- Very low training error
- High testing error
- Poor generalization

Graph:

Error



Model Complexity

Initially both errors decrease.

After a certain point, training error continues decreasing while testing error increases.

6. Possible Reasons for High Testing Error

(i) Overfitting

This is the main reason.

The network has learned every detail of the training data including noise.

Therefore, it cannot predict unseen student scores accurately.

(ii) Small Training Dataset

Suppose only 100 students are used for training.

The model may fail when tested on another school having different students.

More data helps the network learn better patterns.

(iii) Noise in Dataset

Training data may contain incorrect information.

Examples:

- Wrong attendance
- Incorrect marks
- Missing assignment scores

Noise reduces prediction quality.

(iv) Outliers

Some students may have unusual data.

Example:

Student studies only 1 hour but scores 100%.

Such exceptional cases confuse the model.

(v) Too Many Hidden Layers

A very deep network may become unnecessarily complex.

Instead of learning important relationships, it memorizes every training example.

(vi) Lack of Regularization

Without regularization, weights become very large.

Large weights increase overfitting.

(vii) Poor Feature Selection

Irrelevant features reduce model quality.

Example:

Using

- Student shoe size
- Favorite color

These have no relationship with exam scores.

(viii) Improper Train-Test Split

If training data and testing data come from different distributions, the model performs poorly.

Example

Training:

Students from School A

Testing:

Students from School B

Performance decreases.

(ix) Data Imbalance

If most students score between 80–90, the model may fail to predict low-scoring students.

Balanced datasets improve prediction.

(x) Excessive Number of Epochs

Training for too many epochs causes memorization.

Training error decreases,

testing error increases.

7. Bias and Variance Analysis

Machine learning errors are explained using Bias and Variance.

High Bias

- Model too simple
- Underfitting

High Variance

- Model too complex
- Overfitting

Current problem:

Training Error = Low

Testing Error = High

Therefore,

Variance is High

Bias is Low.

8. Corrective Measures**1. Increase Training Data**

Collect more student records.

Larger datasets improve learning.

2. Data Cleaning

Remove

- Missing values
- Duplicate records
- Incorrect marks

Better quality data improves prediction.

3. Feature Selection

Use only relevant features.

Example

- ✓ Attendance
- ✓ Internal marks
- ✓ Assignments
- ✓ Study hours

Remove irrelevant features.

4. Regularization

Use

- L1 Regularization
- L2 Regularization

These reduce large weights and prevent overfitting.

5. Dropout Layer

Randomly deactivate neurons during training.

Example

20–50% neurons are temporarily ignored.

Benefits

- Prevents memorization
 - Improves generalization
-

6. Early Stopping

Stop training once validation error starts increasing.

This prevents overfitting.

7. Cross Validation

Instead of one train-test split,
divide data into multiple folds.

Example

5-Fold Cross Validation

Benefits

- Reliable evaluation
 - Better model selection
-

8. Hyperparameter Tuning

Adjust

- Learning rate
- Batch size
- Number of neurons
- Number of layers
- Epochs

This improves performance.

9. Reduce Network Complexity

Decrease

- Hidden layers
- Number of neurons

A simpler model often performs better on unseen data.

10. Data Normalization

Normalize features between

0 and 1

or

-1 and 1

Training becomes faster and more stable.

9. Performance Evaluation Metrics

Besides MSE, other evaluation metrics include:

Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum |y - \hat{y}|$$

Measures average prediction error.

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{MSE}$$

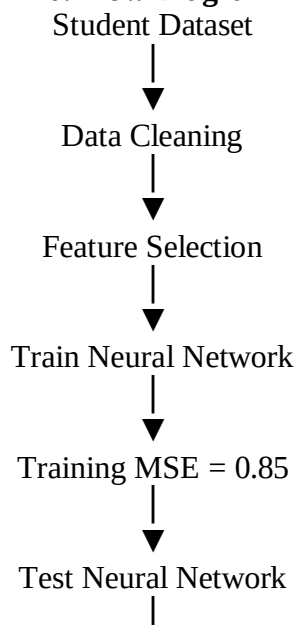
Easy to interpret because it has the same units as the target.

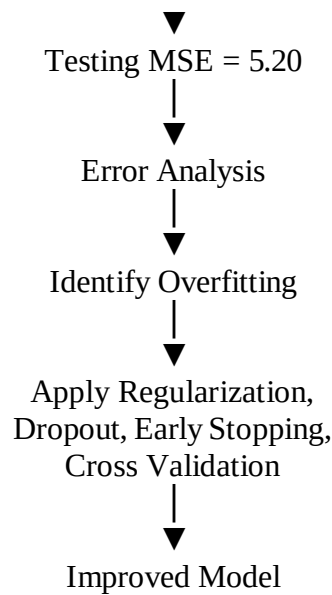
R² Score

Measures how well the model explains the variance in the data.

Higher R² indicates better prediction performance.

10. Flow Diagram





11. Advantages of Error Analysis

- Identifies weaknesses in the neural network.
- Helps detect overfitting and underfitting.
- Improves prediction accuracy.
- Supports better model optimization.
- Increases model reliability.
- Helps select appropriate hyperparameters.
- Enhances performance on unseen data.

12. Limitations

- Requires additional computation time.
- Needs sufficient data for proper analysis.
- Hyperparameter tuning can be time-consuming.
- Large datasets require more memory and processing power.

13. Real-World Example

A university develops a neural network to predict final exam scores using attendance, assignments, internal marks, and study hours.

The model achieves:

- **Training MSE = 0.85**
- **Testing MSE = 5.20**

The large difference shows that the model has memorized the training data rather than learning general patterns. To improve performance, the university collects more student records, removes noisy data, applies dropout and L2 regularization, performs cross-validation, and uses early stopping. These changes reduce the testing error and improve the model's ability to predict scores for new students accurately.

14. Conclusion

The given neural network has a **Training MSE of 0.85** and a **Testing MSE of 5.20**. The significantly higher testing error indicates **overfitting**, where the model learns the training data too closely but fails to generalize to unseen data. Possible reasons include insufficient training data, excessive model complexity, noisy or irrelevant features, and lack of regularization. Corrective measures such as increasing the dataset size, cleaning the data,

applying dropout and L1/L2 regularization, using early stopping, simplifying the network, and performing cross-validation can improve generalization. By applying these techniques, the neural network becomes more accurate, reliable, and capable of predicting student scores effectively on new data.

Code of Conduct:

I M VENKATA SAI(192425223) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M VENKATA SAI

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided